

1. Application Features

1.1 User Signup

- Users can create an account by providing basic information such as name, email, and password, see Figure 2 for User Signup page.

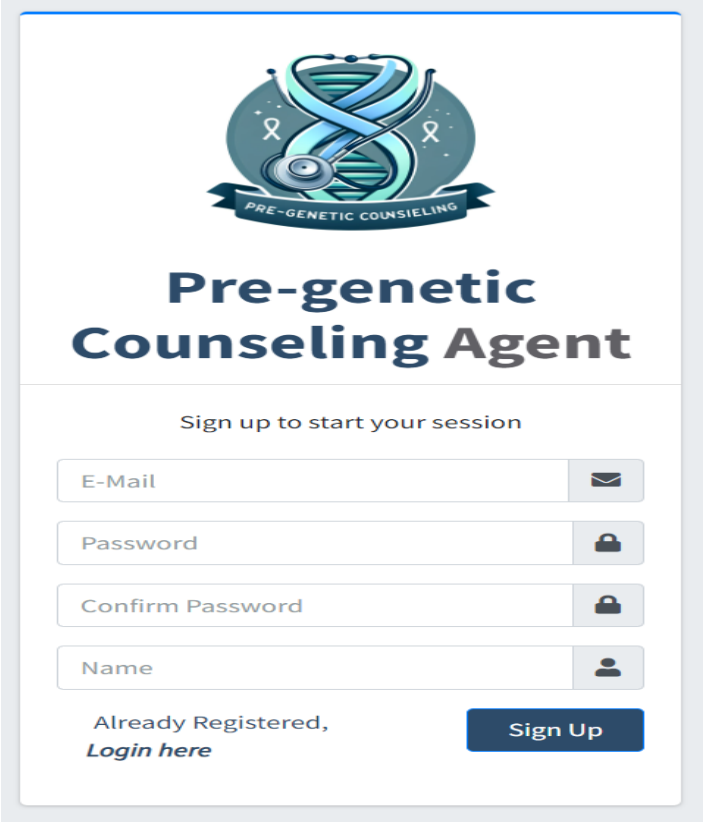
The image shows a user signup page for an application titled "Pre-genetic Counseling Agent". At the top, there is a logo featuring a blue DNA double helix with a stethoscope wrapped around it, set against a circular background with a ribbon banner that reads "PRE-GENETIC COUNSELING". Below the logo, the title "Pre-genetic Counseling Agent" is displayed in a large, bold, dark blue font. Underneath the title, the text "Sign up to start your session" is centered. The form consists of four input fields, each with a corresponding icon on the right: "E-Mail" with an envelope icon, "Password" with a padlock icon, "Confirm Password" with a padlock icon, and "Name" with a person icon. At the bottom left of the form, there is a link that says "Already Registered, Login here". At the bottom right, there is a dark blue button with the text "Sign Up" in white.

Figure 2: User Signup page

1.2 Introductory scripts/Videos

- Users are presented with a series of introductory script/videos explaining the application's purpose and how to use it. Introductory pages include the following:

1.2.1 Starting with welcome page. See Figure 3

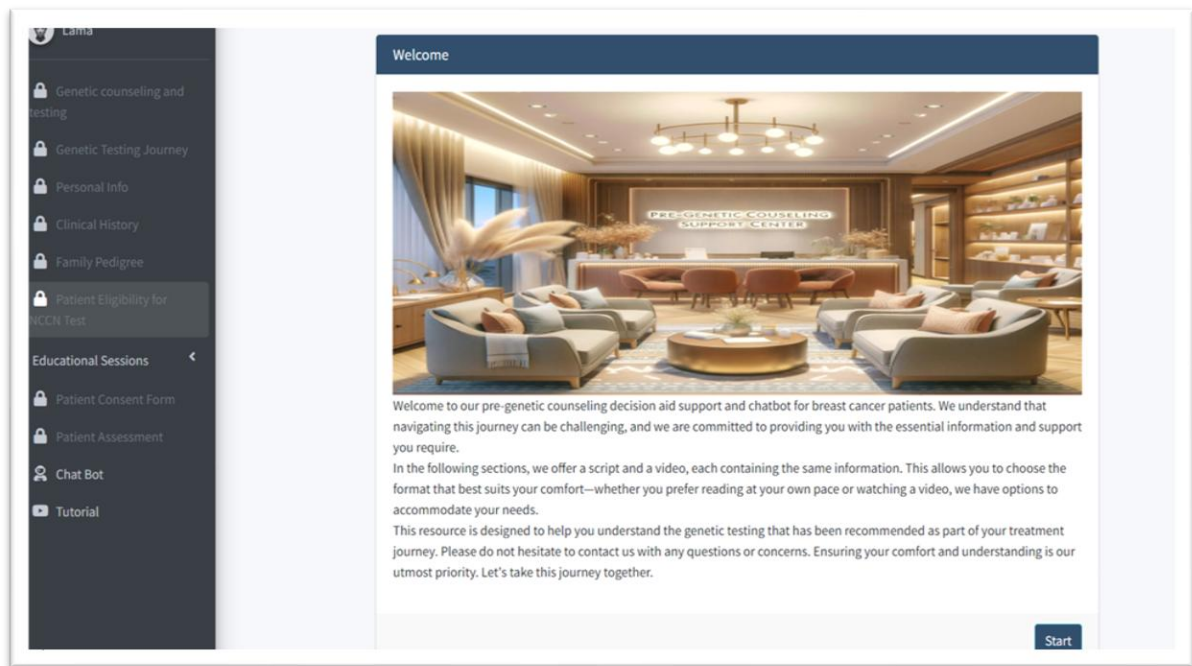


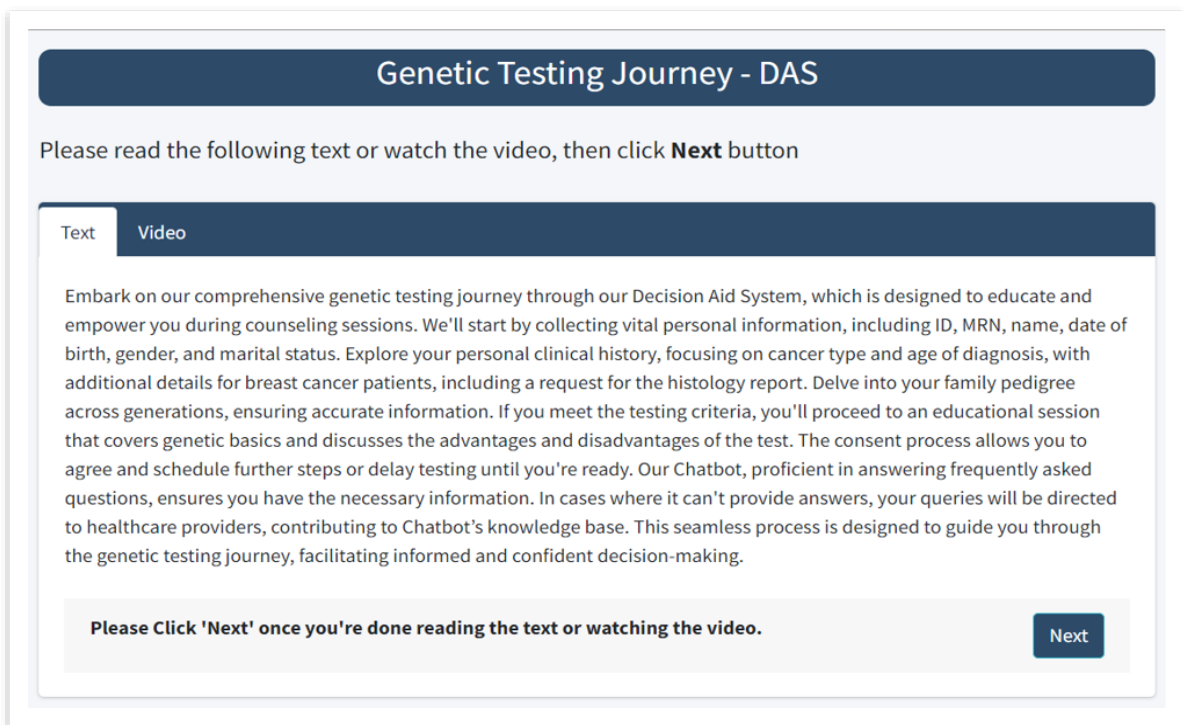
Figure 3: Welcome page

1.2.2 Introduction for Genetic counselling and testing page, presented as script and video. See Figure 4.

Text	Video
"Welcome to our genetic counseling and testing journey, where we embark on a path to unravel the mysteries of our genetic makeup. Genetic counseling serves as a guiding light, offering expert support to individuals and families as they navigate the intricate landscape of gene information. This journey is a profound exploration into understanding genetic risks, making informed decisions, and empowering individuals with the knowledge to shape their health destinies. Genetic testing, a pivotal component of this journey, delves into the very fabric of our DNA, identifying specific changes that hold the key to our unique genetic code. Genetic counseling and testing empower individuals with information and the ability to make choices that align with their genetic profiles. From informed health decisions to the possibility of preventive measures, our genetic journey opens doors to personalized treatments tailored to individual characteristics. The genetic counseling journey unfolds in three main steps: 1. Individuals delve into their medical and family history, learning about genetics, eligibility criteria, and understanding their risk for genetic conditions. 2. Should they undergo testing, they gain insights into the specific procedures and potential outcomes and engage in a consent process. 3. After the test, discussions ensue about the results, decoding their meanings and devising plans based on newfound genetic insights, all with the option for follow-up sessions if needed. These initial steps are guided by a decision-aid system, ensuring individuals have all the information necessary to make educated choices about genetic testing. The actual testing process and the subsequent steps are seamlessly facilitated through the expertise of their physicians. Join us on this transformative journey, where genetic counseling and testing converge to illuminate the path to better health and well-being. Let's embark on a voyage of discovery, where knowledge is power, and our genetic makeup holds the key to informed choices and a healthier tomorrow."	

Figure 4: Introduction for Genetic counselling and testing page.

1.3 Summary for the DAS system framework, presented as script and video. See Figure 5.



Genetic Testing Journey - DAS

Please read the following text or watch the video, then click **Next** button

Text **Video**

Embark on our comprehensive genetic testing journey through our Decision Aid System, which is designed to educate and empower you during counseling sessions. We'll start by collecting vital personal information, including ID, MRN, name, date of birth, gender, and marital status. Explore your personal clinical history, focusing on cancer type and age of diagnosis, with additional details for breast cancer patients, including a request for the histology report. Delve into your family pedigree across generations, ensuring accurate information. If you meet the testing criteria, you'll proceed to an educational session that covers genetic basics and discusses the advantages and disadvantages of the test. The consent process allows you to agree and schedule further steps or delay testing until you're ready. Our Chatbot, proficient in answering frequently asked questions, ensures you have the necessary information. In cases where it can't provide answers, your queries will be directed to healthcare providers, contributing to Chatbot's knowledge base. This seamless process is designed to guide you through the genetic testing journey, facilitating informed and confident decision-making.

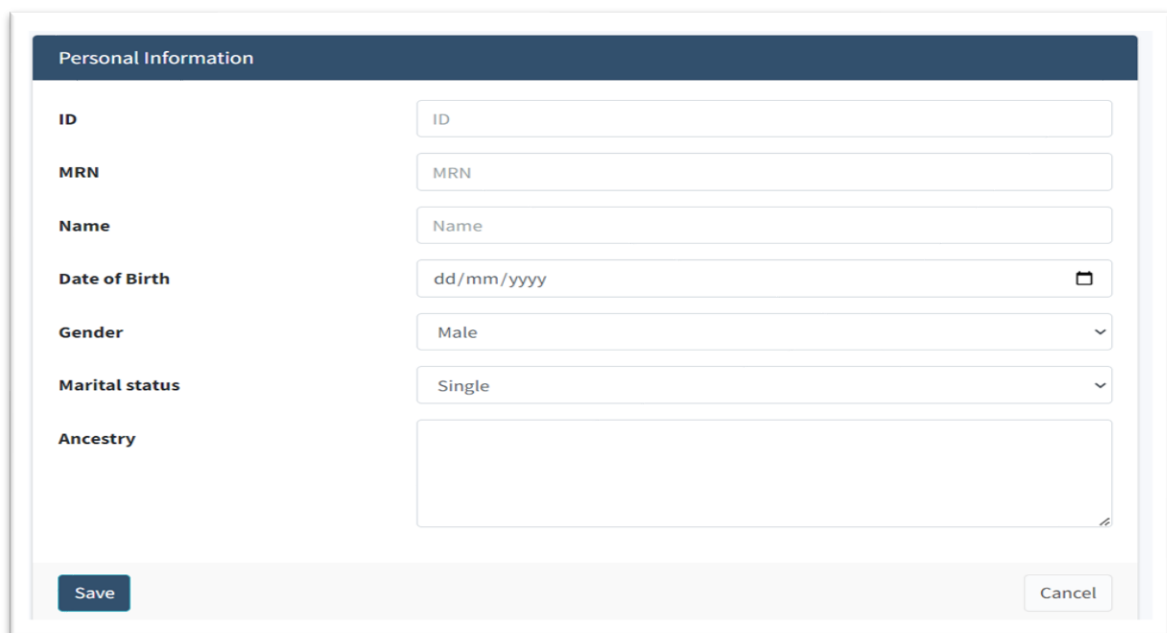
Please Click 'Next' once you're done reading the text or watching the video.

Next

Figure 5: Summary for the DAS system framework-script

1.4 Personal Data Entry

Users input personal information, including age, gender, and medical history, See Figure 6 for personal data entry page.



Personal Information

ID

MRN

Name

Date of Birth 

Gender 

Marital status 

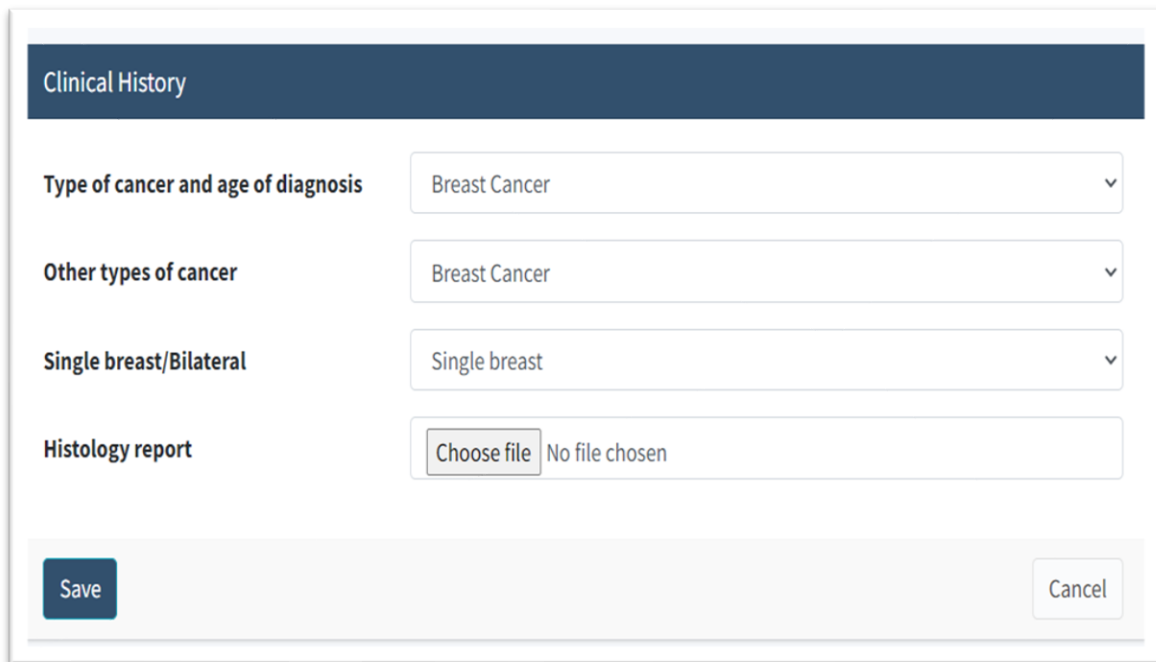
Ancestry

Save **Cancel**

Figure 6: personal data entry page

1.5 Clinical History Input

Users provide detailed clinical history, including a histology report which contains information about tissue samples. See Figure 7 for Clinical History Input page

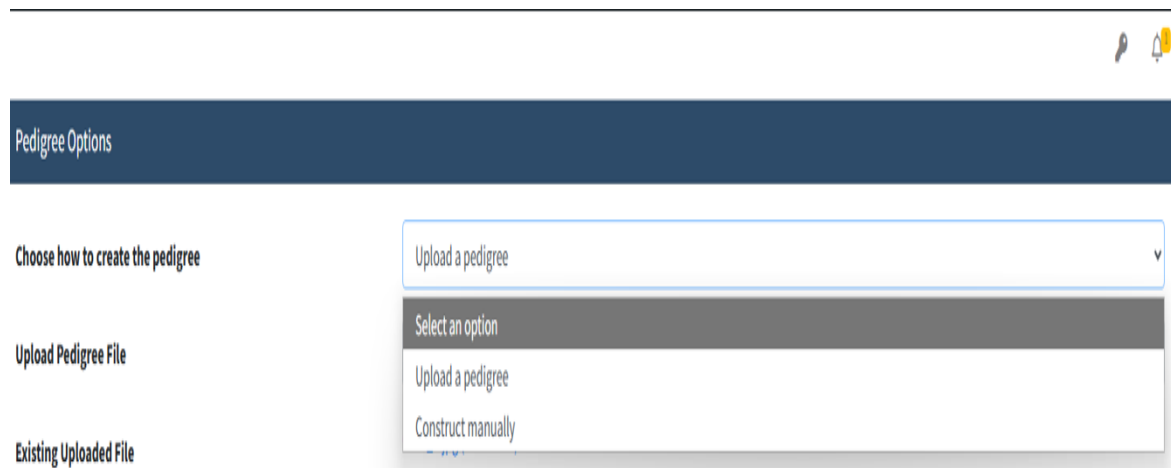


The image shows a web form titled "Clinical History" with a dark blue header. Below the header, there are four input fields. The first three are dropdown menus: "Type of cancer and age of diagnosis" (selected: Breast Cancer), "Other types of cancer" (selected: Breast Cancer), and "Single breast/Bilateral" (selected: Single breast). The fourth field is "Histology report", which contains a "Choose file" button and the text "No file chosen". At the bottom of the form, there are two buttons: "Save" (dark blue) and "Cancel" (light gray).

Figure 7: Clinical History Input page

1.6 Family Pedigree

Users may record family history by either drawing a pedigree in the platform or uploading an existing file. The pedigree is then analysed with AI tools that apply **NCCN** criteria to generate a genetic-risk assessment (see Figures 8a and 8b, Family Pedigree interface).



The image shows a web form titled "Pedigree Options" with a dark blue header. Below the header, there are three input fields. The first is "Choose how to create the pedigree" (dropdown menu, selected: Upload a pedigree). The second is "Upload Pedigree File" (text input field). The third is "Existing Uploaded File" (text input field). A dropdown menu is open below the "Choose how to create the pedigree" field, showing three options: "Select an option" (highlighted), "Upload a pedigree", and "Construct manually".

Figure 8a: Family Pedigree page

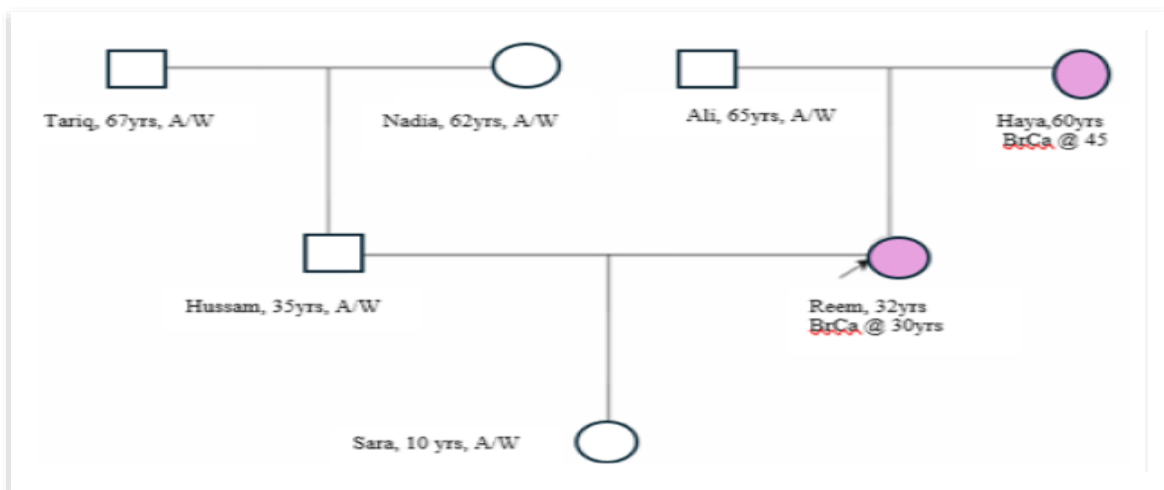


Figure 8b: Family Pedigree/construct manually

1.7 AI-Based Evaluation

The AI analyses the histology report and family pedigree to evaluate patient's eligibility for **NCCN** testing criteria. Results are displayed to the user with detailed explanations. See figure 9 for Patient Eligibility page.

Evaluating Patient Eligibility for NCCN Testing Criteria

Click the "Start" button below to evaluate the patient eligibility for NCCN test

The process will take around 45 seconds

Start

Cancel

Figure 9: Patient Eligibility page

1.8 Educational session

Patients may attend the educational session only after they are confirmed eligible for genetic testing. User will introduce to the educational session scripts /videos that covers:

1.8.1 Concept of genetics and mode of inheritance, See Figure 10 for Concept of genetics and mode of inheritance.

Concept of genetics and mode of inheritance

Please read the following text or watch the video, then click **Next** button

Text

Video

Welcome to our educational journey where we unlock the secrets of DNA and how traits are passed down through generations. Our genetic makeup is like a blueprint for building and maintaining our body. It's composed of DNA, a molecule that contains our genes. Genes are segments of DNA that carry instructions for our development, functioning, growth, and reproduction. These genes are packed into structures called chromosomes, of which humans have 23 pairs, totaling 46.

In autosomal dominant inheritance, only one copy of a mutated gene from one parent is needed to inherit a condition or trait. This means if a parent has the trait, there's a 50% chance it will be passed to each child, regardless of the child's sex. Autosomal recessive inheritance requires two copies of a mutated gene for the individual to exhibit the trait or condition— one from each parent. Parents who each carry one copy of the mutated gene are referred to as carriers. They typically do not show symptoms of the condition. However, there's a 25% chance their child will inherit both copies of the mutated gene and express the condition.

How can detect mutations: In simple terms, sequencing is one of the cutting-edge techniques used to read the exact order of the DNA in our genes. Think of it as decoding the letters in a very long word, where each letter represents a small part of our genetic code. By reading these letters carefully, scientists can identify tiny differences or errors in the genetic code, known as mutations, which can sometimes lead to diseases or affect how we respond to certain medications.

Genetics is a key to unlocking many mysteries of our health and traits. By understanding how traits and conditions are passed down, we can better manage our health, make informed decisions, and support those affected by genetic conditions.

Please Click 'Next' once you're done reading the text or watching the video.

Next

Figure 10: Concept of genetics and mode of inheritance.

1.8.2 Multigene panel and Outcome results. See figure 11 for multigene panel and outcome results.

Multigene panel and Outcome results

Please read the following text or watch the video, then click **Next** button

Text

Video

As you embark on your genetic testing journey, there are various options available, including targeted testing for a single gene, a selection of genes related to breast cancer risk, or a more extensive panel that screens for a variety of cancer risks. In this particular stage, your testing will encompass a multigene panel, focusing on genes with a high likelihood of increasing breast cancer susceptibility. The specific genes included in this testing are BRCA1, BRCA2, CDH1, PALB2, PTEN, and TP53.

Potential outcomes of the genetic testing can include positive, negative, uncertain, and incidental findings:

- Positive – The test has identified a mutation correlated with a heightened risk of hereditary cancer. This knowledge could enable you and your healthcare provider to make informed decisions regarding your medical care, including enhanced screening protocols, risk-lowering surgeries, and preventive pharmacological measures.
- Negative – No mutations have been detected in the genes analyzed during your test. This signifies that:
 - As the initial family member is undergoing testing, your baseline risk for cancer parallels that of the average person. However, there remains a possibility of an undetected hereditary cancer risk, which this test may not capture, whether within the tested gene(s) or another gene associated with hereditary cancer.
 - If your results are negative for a familial mutation, your genetic risk may be considered equivalent to the general population.
- Uncertain – A genetic variation was found; however, its association with cancer risk is currently unknown. Your cancer risk remains at least on par with the general populace, and there could be an above-average risk due to this genetic variation or another undetected predisposition related to the tested gene(s) or a different gene involved in hereditary cancer.
- Incidental Findings: Additional findings not specifically related to the assessed cancer risk may emerge during the analysis. These incidental findings could have health implications and require further investigation or medical follow-up.

Figure 11: Multigene panel and Outcome results.

1.8.3 Benefits, Risks, and Limitations of genetic testing. See Figure 12 for Benefits, Risks, and Limitations of the genetic testing.

Benefits, Risks, and Limitations of the genetic testing

Please read the following text or watch the video, then click **Next** button

Text

Video

- Benefits of the genetic testing: Genetic test results help you and your doctor make more informed choices about your health care, such as screening, risk-reducing surgeries, and preventive medication strategies. Identifying gene mutation(s) in a family enables other blood relatives to determine whether or not they share the same hereditary cancer risks.

If you are positive, you should discuss with your Physician / Counselor how hereditary cancer is inherited and learn about the chance your children and blood relatives may have inherited the same mutation(s) in the gene(s) tested.

Understanding your risk for hereditary conditions empowers you to make informed decisions about preventive measures and personalized healthcare. This newfound awareness provides psychological relief, easing anxieties or presenting early detection opportunities for proactive health management. Genetic testing enhances your understanding of your genetic makeup, offering profound insights into potential health outcomes. It supports healthcare professionals in tailoring treatment plans to your unique genetic profile.

Suppose you test negative for a known mutation in your family. In that case, you cannot pass on that mutation to your children, and you may be considered to have the same genetic risks for cancer as others in the general population.

- Risks of the genetic testing: Genetic testing requires DNA, most often provided from a sample of blood from a saliva sample or a tumor sample. Side effects of having blood drawn are uncommon but may include dizziness, fainting, soreness, bleeding, bruising, and rarely, infection.

You may experience heightened anxiety or emotional distress, particularly if results indicate an increased risk. Challenges may

Figure 12: Benefits, Risks, and Limitations of the genetic testing.

1.9 Consent form and patient's assessments:

Once a patient chooses to undergo genetic testing, they can electronically sign the consent form and complete a 10–15-item questionnaire to assess their understanding of the procedure.

1.10 ChatBot

This OpenAI-powered chatbot, enriched with **NCCN** guideline content and other Frequently Asked Questions, can answer questions across the domains illustrated in figure 13. Patients navigate the application and interact via the Chatbot page, see figure 14.

Chatbot for pre-genetic counseling session:

I can help with the following type of information:

1. FAQs for High-Risk Breast Cancer Patients in pre-genetic counseling
 - a. Personal and Family History pre-genetic counseling.
 - b. Psychological pre-genetic counselling.
 - c. Educational pre-genetic counselling.
 - d. Clinical pre-genetic counselling.
 - e. Sociological pre-genetic counselling.
 - f. FAQs on high-penetrance breast cancer susceptibility genes: BRCA1, BRCA2, CDH1, PALB2, PTEN, TP53
2. Gene summary: Risk and Management
3. Genetic /Familial high-risk assessment and counseling
4. Multi-Gene testing
5. High penetrance Breast cancer/Ovarian Cancer Susceptibility Genes
6. BRCA related Breast/Ovarian cancer syndrome
7. Cancer risk management and screening
8. Pathogenetic/ likely pathogenic variants associated with Breast/Ovarian cancer
9. Genetic testing criteria

Figure 13: Information Domains Covered by the Pre-Genetic Counselling Chatbot

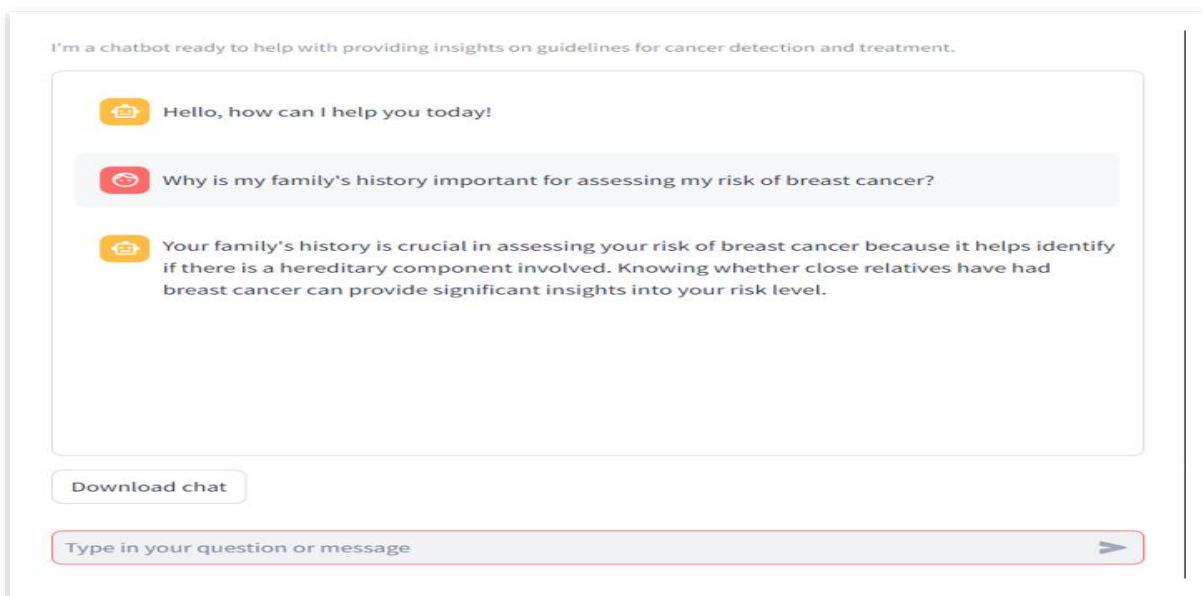


Figure 14: ChatBot page